

SHIVAM SINGH

AI Researcher | Vision-Language Models | Multimodal Reasoning | Embodied AI

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SUMMARY

AI Researcher with **3+ years** of research experience working on Vision-Language Models, multimodal reasoning, video understanding, and embodied AI. Developed multimodal foundation models and video reasoning systems, built scalable multimodal data pipelines, and worked on LLM-based planning, task anticipation, and human-robot collaboration.

EXPERIENCE

pixelstwo Sep 2026 – Present
Collaboration Remote

Focus: Video Data Quality & Real-to-Sim / Sim-to-Real Systems

- Performing **quality control of video data**, ensuring recordings are reliable and consistent for downstream training and evaluation.
- Working on **real-to-sim and sim-to-real systems**.

intuition.dev Aug 2026 – Sep 2026
Collaboration Remote

Focus: Real-World Video Pipelines

- Label-free Segmentation and Grounded Description of Egocentric Work Videos** 2026
 - Built a **geometry-first, label-free pipeline** that converts raw body-camera recordings of manual work into **clean, standalone sub-task clips**: removed inactive/noisy footage with classical detectors and detected **task-transition points** from metric camera trajectories using purely classical image-processing heuristics, without relying on large models for segmentation and prior annotations.
 - Structured raw videos for downstream **VLM-based annotation** by isolating relevant task segments before captioning, **reducing VLM inference cost** and enabling more consistent, task-level annotations under a fixed JSON schema with grounded actions. [Technical Report]

AMD India Private Limited Jun 2025 – Jul 2026
Research Fellow AI Models & Agents Team

Focus: VLMs, Video Reasoning & Spatial Understanding

Supervisors: Saptarshi Majumder, Pratik Prabhanjan Brahma

- Spatial Video Understanding from Raw RGB Videos** 2026
 - Proposed **SlotMem**, a **memory-augmented** transformer framework for object tracking, spatial understanding, and long-horizon video reasoning from raw monocular RGB videos.
 - Developed a **scalable pipeline** combining Depth-Anything-3, YOLO-World, and Vision-Language Models to reconstruct 3D scenes, estimate camera trajectories, and generate supervision for spatial, temporal, and goal-oriented reasoning.
 - Trained SlotMem using our automatically generated supervision together with diverse videos from VLM-3R and VICA-322K, achieving **state-of-the-art** performance on OST-Bench (**44.4%**) and outperforming all prior baselines.

EDUCATION

IIIT Hyderabad
M.S.
Electronics and Communication Engineering
2023 – 2026 CGPA: 8.33/10

RKIT, Ghaziabad
B.Tech
Electronics and Communication Engineering
2017 – 2021 CGPA: 8.07/10

SKILLS

Research Areas
LLMs, VLMs, VLAs, Reasoning Models, Computer Vision, Multimodal Learning, Agentic AI, Embodied AI

Training & Inference
PyTorch, Hugging Face, LLaMA-Factory, SFT, RLHF, LoRA, DeepSpeed, vLLM, ROCm

Data & Infrastructure
Multimodal Data Curation, Synthetic Data, Benchmarking, Docker, Slurm, OpenCV, Open3D

Programming & Robotics
Python, C++, C, PDDL, RDDL, ROS, Gazebo, AI2-THOR

RESEARCH HIGHLIGHTS

- 3** first/co-first author papers at **A*** robotics venues (**ICRA 2024, ICRA 2025, IROS 2026**)
- Built a compact **4B** VLM for video-grounded spatial understanding, matching a **59×** larger model
- Achieved **SOTA** on OST-Bench (**44.4%**) through improved spatial reasoning and memory
- Built scalable **multimodal data curation pipelines**, from structuring raw real-world work videos to generating grounded VLM training data
- Trained multimodal models on **8×** AMD Instinct MI325 GPUs and deployed on-device on AMD Ryzen AI

• Pause and Think: Video-Grounded Assistive Action Suggestion IROS 2026

- Proposed a reasoning-centric framework for video-grounded assistive action suggestion by introducing *Pause-and-Think-T* (~10k structured reasoning QA pairs) with explicit reasoning supervision and *Pause-and-Think-B*, a **300-sample** benchmark for grounded contextual understanding and goal-oriented planning across Epic-Kitchens, Assembly101, and Ego4D.
- Designed a scalable data generation pipeline with annotation refinement using GPT-OSS-120B, goal-oriented clip segmentation (5–20s), and structured `<think>/<answer>` QA generation using Qwen3-VL-235B with majority voting and self-consistency verification.
- Fine-tuned a compact **4B** transformer-based VLM using LLaMA-Factory and DeepSpeed on 8×AMD Instinct MI325 GPUs, served it with vLLM, and achieved **58.0%** accuracy, matching Qwen3-VL-235B (**58.9%**, **59× larger**) and GPT-5.2 on scene understanding, outperforming GPT-4o, with strong zero-shot generalization on EgoThink and TempCompass without benchmark-specific training.
- Deployed the optimized model on an **AMD Ryzen AI** platform for real-time on-device inference; as **First Author**, led the research, algorithm design, dataset creation, model training, evaluation, deployment, and manuscript preparation. [[Code](#) | [Paper](#)]

IIIT Hyderabad

M.S. Researcher

Focus: Embodied AI & Robot Planning

Supervisor: [Prof. Madhava Krishna](#)

Collaborators: [Dr. Mohan Sridharan](#), [Dr. Krishna Murthy Jatavallabhula](#), [Dr. Brojeshwar Bhowmick](#)

Jan 2023 – May 2025

[Robotics Research Centre](#)

• Hybrid Planning with LLMs & Probabilistic Models ECMR 2025

Mentor: Guided framework design and ablation experiments for task planning under uncertainty with probabilistic effects in multi-agent settings. [[Paper](#)]

• LLM & Knowledge Graphs for Task Decomposition IEEE ICRA 2025

Project Lead: Designed a framework that integrates Large Language Models (LLMs), Knowledge Graphs (KGs), and Human-in-the-Loop (HITL) feedback to enable embodied agents to perform long-horizon planning for unseen tasks and **progressively adapt** to new environments. [[Website](#) | [Paper](#)]

• Task Anticipation for Multi-Agent Systems ICRA & RSS 2024 Workshops

Project Lead: Extended LLM-based task anticipation to long-horizon planning in **multi-agent settings**, enabling robotic agents to anticipate domain-specific tasks, adapt to unexpected changes in human behavior, and handle environmental uncertainties for effective human–robot collaboration. [[Website](#) | [Paper](#)]

• Task Anticipation using LLMs IEEE ICRA 2024

Co-author & Project Lead: Developed an intelligent household agent that performs **long-horizon planning** by anticipating user actions through learning task execution patterns from user behavior and preferences, leveraging Large Language Models (LLMs) and Planning Domain Definition Language (PDDL)-based task and action representations. [[Website](#) | [Paper](#)]

AlsoEnergy

Technical Support Engineer

Apr 2021 – Dec 2021

India

- PowerTrack platform support for commercial solar clients: troubleshooting, data visualization, hardware integration.

PUBLICATIONS

- [Pause and Think: A Dataset and Benchmark for Video-Grounded Assistive Action Suggestion](#). Shivam Singh, S Majumder, P Prabhanjan, Z Liu, E Barsoum. **IEEE/RSJ IROS 2026 (A*)**. Work done at AMD.
- [Anticipate, Adapt, Act: A Hybrid Framework for Task Planning](#). N Dash, A Kaura, Shivam Singh, R Singh, S Banerjee, M Sridharan, M Krishna. **IEEE ECMR 2025**.
- [AdaptBot: Combining LLM with Knowledge Graphs and Human Input for Generic-to-Specific Task Decomposition and Knowledge Refinement](#). Shivam Singh, K Swaminathan, N Dash, R Singh, S Banerjee, M Sridharan, M Krishna. **IEEE ICRA 2025 (A*)**, Atlanta, US.
- [Anticipate & Collab: Data-driven Task Anticipation and Knowledge-driven Planning for Human-robot Collaboration](#). Shivam Singh*, K Swaminathan*, R Arora*, D Das, S Banerjee, M Sridharan, M Krishna. **RSS 2024 Workshop** on Task Specification for General-Purpose Intelligent Robots, Delft, Netherlands.
- [Anticipate & Act: Integrating LLMs and Classical Planning for Efficient Task Execution in Household Environments](#). R Arora*, Shivam Singh*, K Swaminathan, A Datta, S Banerjee, B Bhowmick, KM Jatavallabhula, M Sridharan, M Krishna. **IEEE ICRA 2024 (A*)**, Yokohama, Japan.
- [DaTAPlan: Data-driven Task Anticipation and Knowledge-driven Planning for Human-robot Collaboration](#). Shivam Singh*, K Swaminathan*, R Arora*, R Singh, A Datta, D Das, S Banerjee, M Sridharan, M Krishna. **ICRA 2024 Workshop** on Cooking Robotics: Perception and Motion Planning, Yokohama, Japan.

* equal contribution

PATENTS

[Systems and Methods for Data-driven Task Anticipation and Knowledge-driven Planning for Human-Robot Collaboration](#)

Shivam Singh, K Swaminathan, R Arora, R Singh, A Datta, D Das, S Banerjee, M Sridharan, M Krishna.

US Patent App. 19089689

Published Nov 6, 2025